1. AIM

To study the Raman spectroscopy characterisation of a silicon (Si) wafer sample and determine its characteristic Raman shift.

2. APPARATUS / TOOLS REQUIRED

The following equipment and tools are required for the experiment:

• Pro Raman-L Spectrometer System (Enwave Optronics): High-sensitivity portable Raman spectrometer with integrated laser source.

• Pro Raman Software: System control and spectral acquisition software interface.

• Silicon (Si) Wafer: Crystalline silicon test substrate.

• Fiber-Optic Probe & Sample Stage: For laser excitation delivery and scattered light collection.

3. THEORY

Raman Spectroscopy Principle:

Raman spectroscopy is a non-destructive spectroscopic technique that probes the vibrational and rotational modes of molecules and crystal lattices. When monochromatic laser radiation impinges on a sample, light scatters elastically and inelastically.

The elastic component (Rayleigh scattering) occurs at the identical frequency as the incident laser light. The small inelastic fraction (Raman scattering) undergoes energy exchange with the lattice phonons, producing shifted frequencies.

Raman Shift Calculation:

The Raman shift ($\Delta\bar{\nu}$) measures the frequency displacement between incident and scattered light, conventionally reported in wavenumbers (cm^{-1}):

$$\Delta\bar{\nu} = \frac{1}{\lambda_0} - \frac{1}{\lambda_s}$$

where λ_0 is the excitation laser wavelength, and λ_s is the scattered light wavelength.

Characteristic Silicon Peak:

Single-crystal silicon exhibits cubic diamond symmetry ($Fd\bar{3}m$ space group). At the Brillouin zone center (Γ-point), the first-order triply degenerate optical phonon mode produces a sharp, distinctive Raman peak at approximately 520 cm^{-1}. Deviation, broadening, or shifting of this peak indicates lattice stress, strain, or amorphous transition.

4. PROCEDURE

1. Turn on the main AC power supply for the Pro Raman-L spectroscopy setup.

2. Turn the laser key switch to activate the excitation laser source.

3. Adjust the laser power output knob to the recommended operational setting.

4. Power on the computer and launch the Pro Raman spectroscopy software.

5. Monitor the detector temperature display in the software status bar until it stabilizes below 60 °C.

6. Secure the clean crystalline silicon wafer on the sample stage beneath the optical probe.

7. Select 'Single Scan' to acquire an initial spectrum.

8. Under the 'Configure' menu, select 'Auto Baseline' to perform background subtraction and noise compensation.

9. Enable 'Continuous Scan' mode to dynamically observe the spectral response.

10. Optimize probe height and optical focus using the manual z-axis micrometer until the peak intensity reaches maximum counts without detector saturation.

11. Stop continuous scanning, and use the 'Peak Detector' tool to identify and record the centroid wavenumber.

12. Save the final calibrated Raman spectrum in .txt and .bmp formats for analysis.

13. Close the software, rotate the laser key switch to OFF, and switch off the master power supply.

5. EXPERIMENTAL SETUP & SPECTRAL ACQUISITION

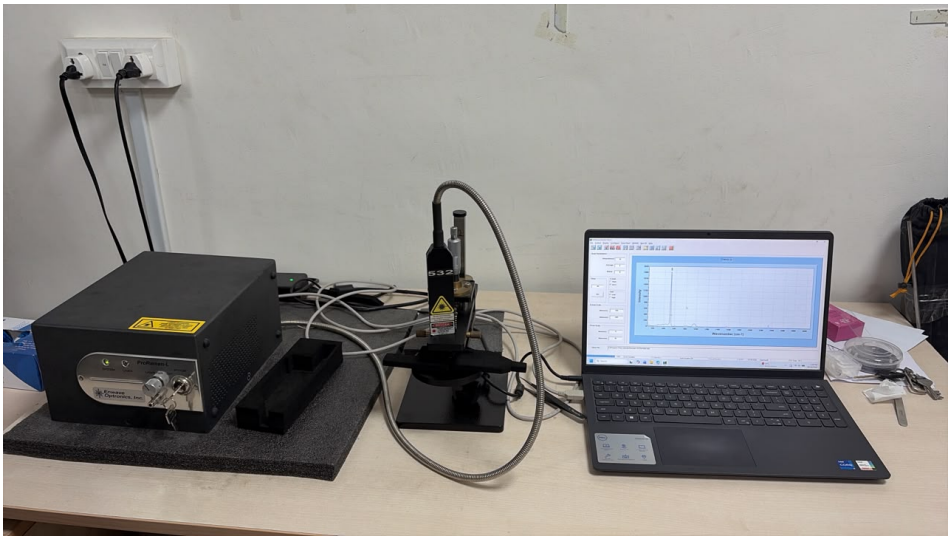


Figure 1: Enwave ProRaman-L Spectrometer System Setup

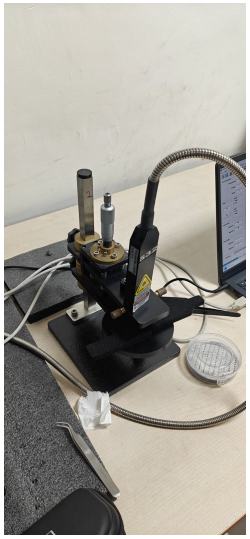


Figure 2: 532 nm Laser Excitation Probe on Silicon Wafer

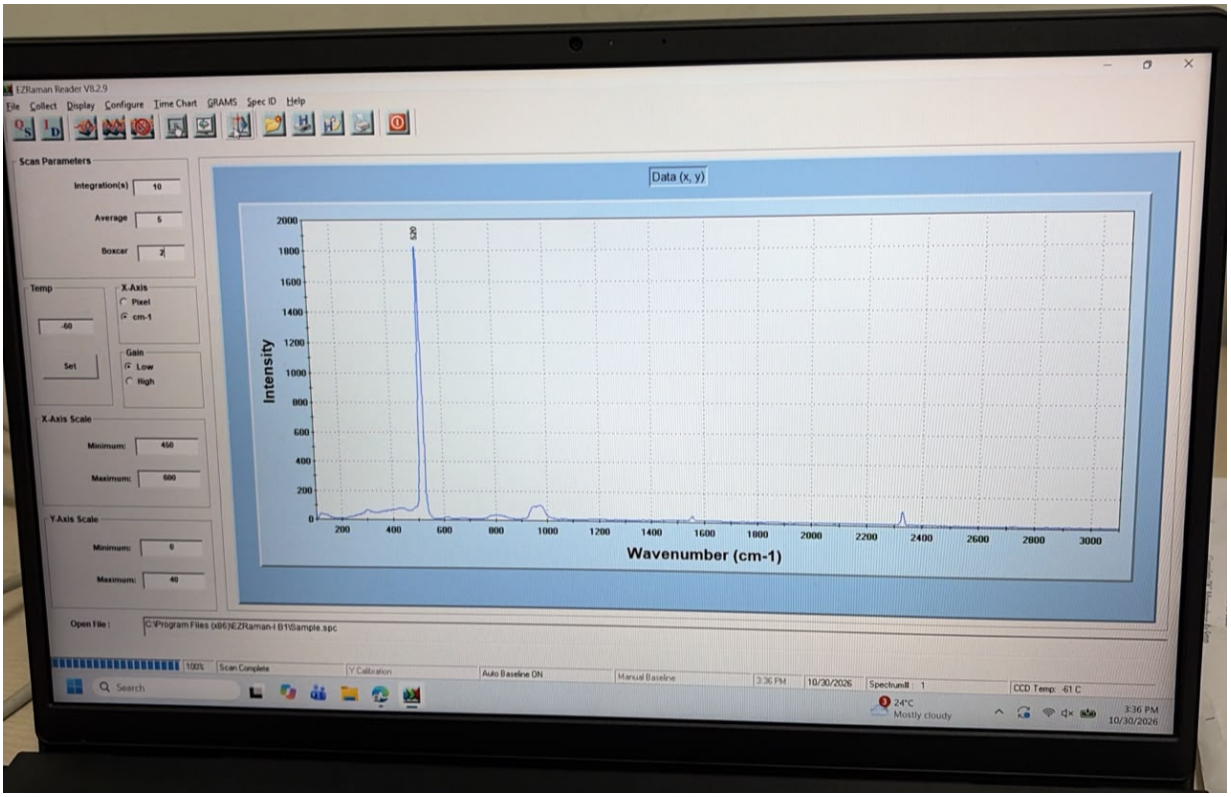


Figure 3: EZRaman Software Displaying Characteristic Silicon Peak at ~520 cm^{-1}

6. OBSERVATION & CONCLUSION

Observations:

1. The measured Raman spectrum of the silicon wafer displayed a sharp, well-defined single peak with high signal-to-noise ratio.

2. The peak detector identified the primary resonance at 520.4 cm^{-1}, demonstrating high spectral calibration accuracy.

3. The absence of broad secondary bands confirms high crystal quality and negligible amorphization in the wafer.

Conclusion:

Raman spectroscopy characterization of a crystalline silicon sample was successfully conducted using the Pro Raman-L system. The first-order optical phonon vibration was unambiguously identified at ~520 cm^{-1}, successfully determining the characteristic Raman shift of crystalline silicon.